

Known at the origin o

Three orthogonal blocks

Error-weighted handover

Output

Unit's own history

hourly load, temperature
and calendar before o



pseudo-origins o – 744k



completed forecasts give every
candidate its own error record

TimesFM 2.5

zero-shot, load only
1,344 h $\rightarrow$ 744 h, issued at
o and at o – 744k ($k \leq 6$)

**Level ℓ**

six historical candidates: recent, long-history
and annual means, with and without weather

+ TimesFM window mean

weights $\propto 1/\text{MSE}$ at pseudo-origins, shrunk to equal ($K_0 = 8$)

**Centred daily path b**

seven historical candidate paths: 8-, 4- and 2-week
day types, the annual window, the weather response

same error weighting ($K_0 = 2$); $\sum_d b_d = 0$

**Within-day shape w**

day-demeaned TimesFM trajectory w^T

+ the unit's analog-day shape S : same day type,
 ± 14 days of year, up to 8 past days

$w = w^T + \omega_k (S - w^T)$, $\omega_k \leq 1/2$ from the unit's errors
at 3 pseudo-origins, shrunk to 0 (2.0)

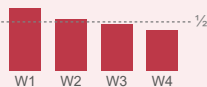
**Week-by-week
handover**

$$m_d = (1 - \alpha_w) m_d^H + \alpha_w m_d^T$$

m^H : history ($\ell + b$)

m^T : TimesFM daily means

α_w : least squares on ≤ 6
pseudo-origin pairs (fixed
division vs TimesFM),
shrunk to $1/2$ ($K_0 = 2$)

**ANKYRA**

$$F_{d,h} = m_d + w_{d,h}$$

projected onto $F \geq 0$

no training on the
target series

**off-state switch**

last 168 h $\leq 10^{-6}$ kW
 $\rightarrow$ TimesFM unchanged

Readouts

energy 744 ℓ

peak $\max_d (\hat{L}_d + A_{T_d})$

interval pseudo-origin
residual quantiles

Exact identity: $\text{MSE} = \ell(e)^2 + \langle b_d(e)^2 \rangle_d + \langle w_{d,h}(e)^2 \rangle_{d,h}$ — the blocks are orthogonal, so each source is scored and anchored block by block (all three since 2.0).

from the unit's own history from TimesFM 2.5 ANKYRA's error-weighted handovers